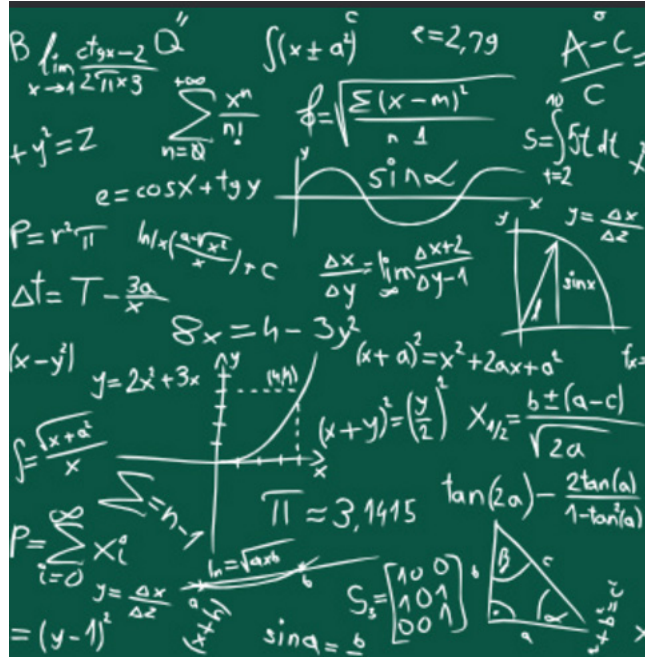


Trigonometric functions



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1 Trigonometric functions

Definition 1: sine function and cosine function

The trigonometric functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \sin x \quad (1)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \cos x \quad (2)$$

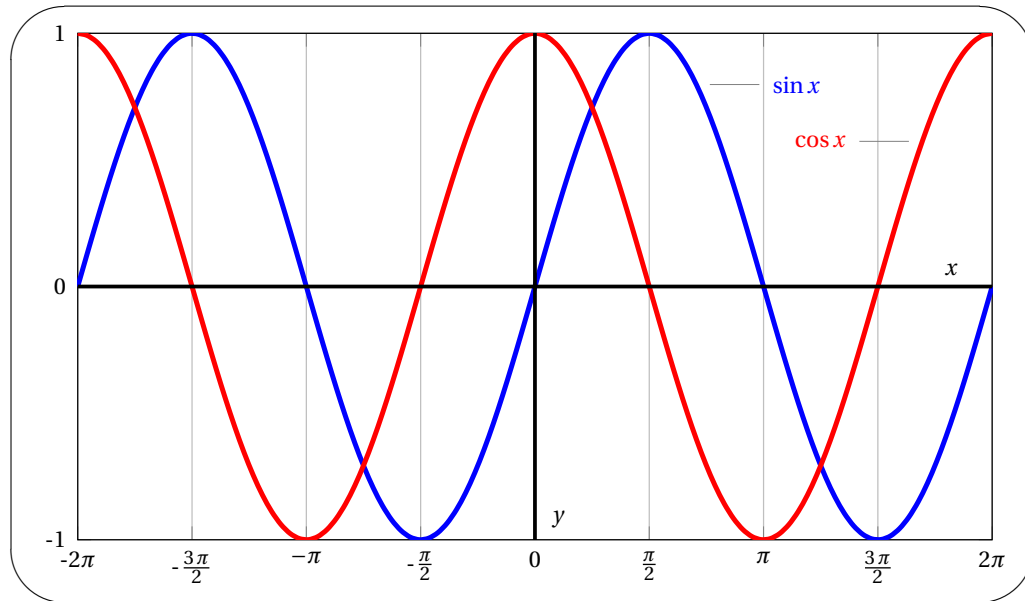
are called **sine** and **cosine**.

- The sine is an **odd** function, while the cosine is an **even** function. Hence we have:

$$\sin(-x) = -\sin x, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \cos(-x) = \cos x, \quad \forall x \in \mathbb{R}$$

- The sine and the cosine are periodic functions with period:

$$T = 2\pi \quad \text{and we have the fundamental identity} \quad \sin^2 x + \cos^2 x = 1, \quad \forall x \in \mathbb{R}$$



We have

$$\cos\left(x - \frac{\pi}{2}\right) = \sin x, \quad \forall x \in \mathbb{R}$$

- From the sign and the monotonicity of the sine function we have:

$$\sin x = 0 \iff x = k\pi, \quad \forall k \in \mathbb{Z} \quad \begin{cases} \sin x > 0 & \text{if } x \in (2k\pi, 2k\pi + \pi), \quad \forall k \in \mathbb{Z} \\ \sin x < 0 & \text{if } x \in (2k\pi - \pi, 2k\pi), \quad \forall k \in \mathbb{Z} \end{cases}$$

$$\begin{cases} \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

- From the sign and the monotonicity of the cosine function we have:

$$\cos x = 0 \iff x = k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z} \quad \begin{cases} \cos x > 0 & \text{if } x \in (2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}), \forall k \in \mathbb{Z} \\ \cos x < 0 & \text{if } x \in (2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}), \forall k \in \mathbb{Z} \end{cases}$$

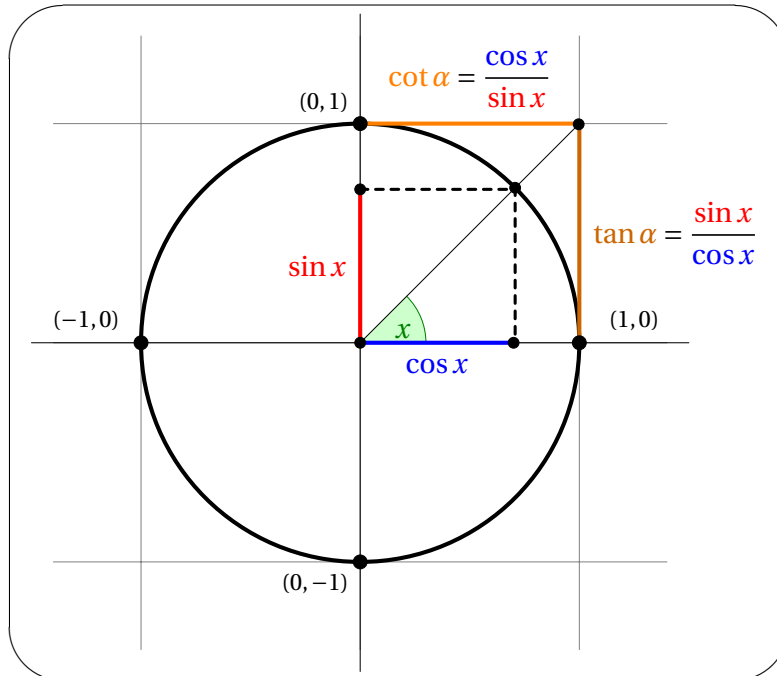
$$\begin{cases} \text{if } x \in (2k\pi + \pi, 2k\pi + 2\pi), \forall k \in \mathbb{Z} & f \text{ is increasing} \\ \text{if } x \in (2k\pi, 2k\pi + \pi), \forall k \in \mathbb{Z} & f \text{ is decreasing} \end{cases}$$

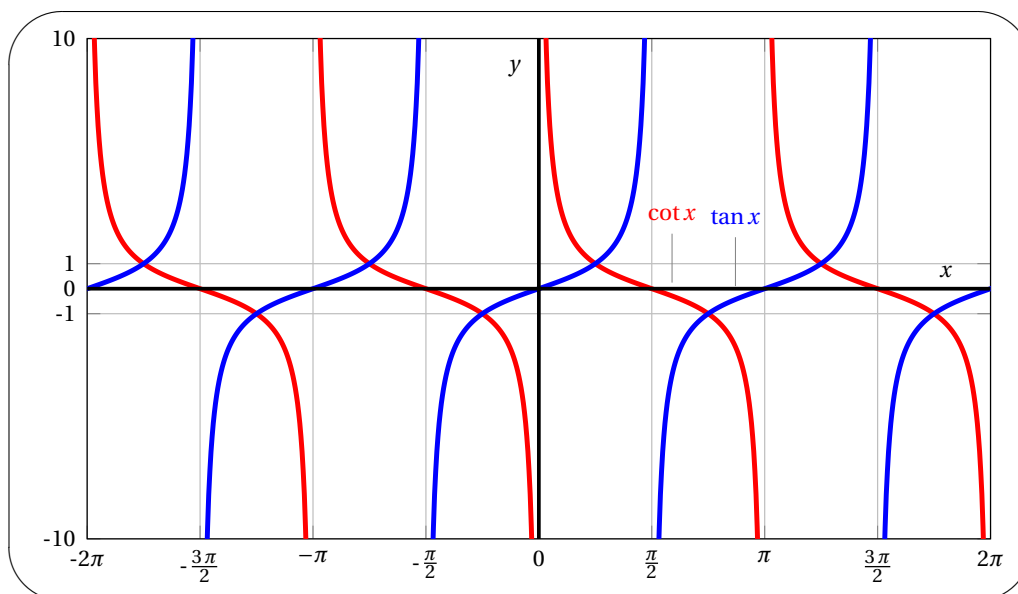
The trigonometric functions:

$$f: \mathbb{R} \setminus \{k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \tan x = \frac{\sin x}{\cos x} \quad (3)$$

$$f: \mathbb{R} \setminus \{k\pi, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \cot x = \frac{\cos x}{\sin x} \quad (4)$$

are called **tangent** and **cotangent**; they are both odd and periodic with period $T = \pi$.



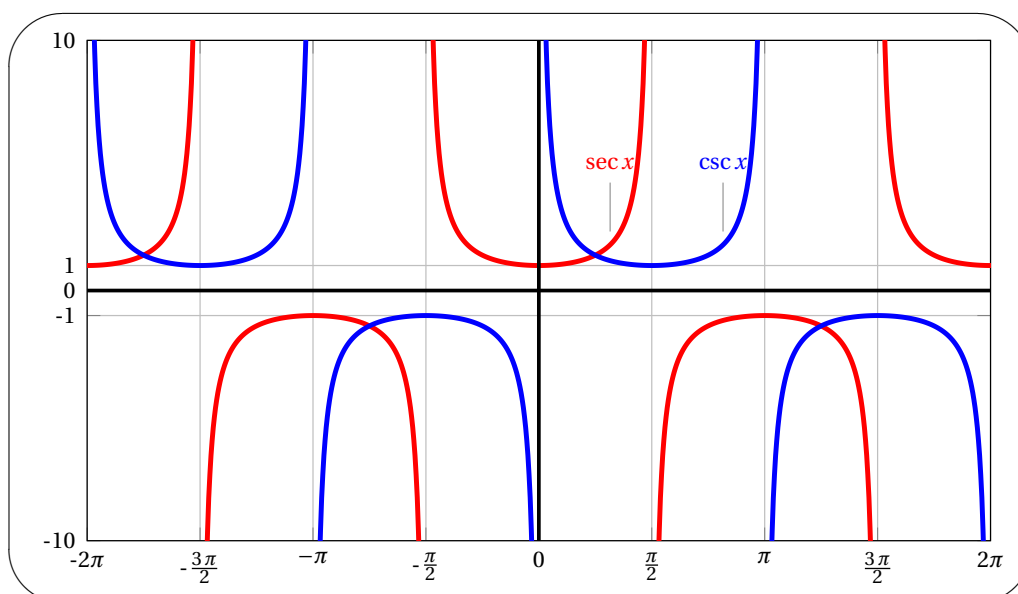


The trigonometric functions:

$$f: \mathbb{R} \setminus \{k\pi - \frac{\pi}{2}, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \sec x = \frac{1}{\cos x} \quad (5)$$

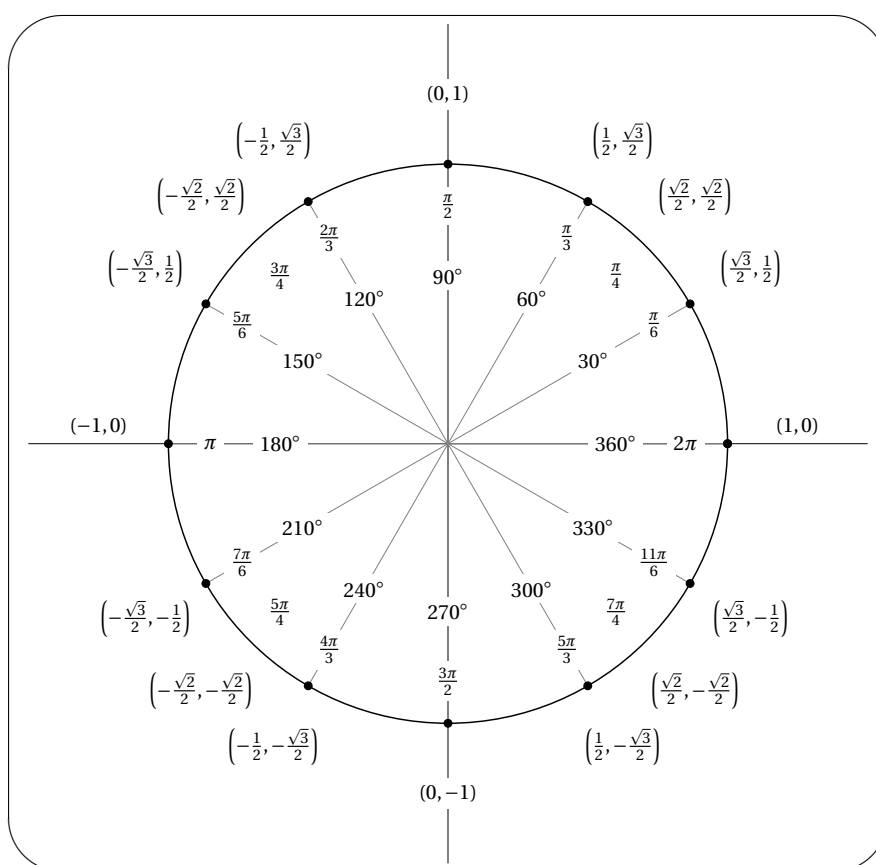
$$f: \mathbb{R} \setminus \{k\pi, \forall k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad f: x \mapsto \csc x = \frac{1}{\sin x} \quad (6)$$

are called **secant** and **cosecant** and they are periodic with period $T = 2\pi$. The secant function is even, while the cosecant function is odd.



2 Values of the trigonometric functions

	cos	sin	tan	cot	sec	csc
0	1	0	0	$\pm\infty$	1	$\pm\infty$
$x = \frac{\pi}{6}$ (30°)	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$	$\frac{2}{3}\sqrt{3}$	2
$x = \frac{\pi}{4}$ (45°)	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1	$\sqrt{2}$	$\sqrt{2}$
$x = \frac{\pi}{3}$ (60°)	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2}{3}\sqrt{3}$
$x = \frac{\pi}{2}$ (90°)	0	1	$\pm\infty$	0	$\pm\infty$	1
$x = \pi$ (180°)	-1	0	0	$\pm\infty$	-1	$\pm\infty$



	cos	sin	tan	cot
x	$\cos x$	$\sin x$	$\tan x$	$\cot x$
$-x$	$\cos x$	$-\sin x$	$-\tan x$	$-\cot x$
$\frac{\pi}{2} + x$	$-\sin x$	$\cos x$	$-\cot x$	$-\tan x$
$\frac{\pi}{2} - x$	$\sin x$	$\cos x$	$\cot x$	$\tan x$
$\pi + x$	$-\cos x$	$-\sin x$	$\tan x$	$\cot x$
$\pi - x$	$-\cos x$	$\sin x$	$-\tan x$	$-\cot x$

3 Main trigonometric formulas

Relations among the trigonometric functions

	$\sin x$	$\cos x$	$\tan x$	$\cot x$
$\sin x$	-	$\pm \sqrt{1 - \cos^2 x}$	$\pm \sqrt{\frac{\tan^2 x}{1 + \tan^2 x}}$	$\pm \sqrt{\frac{1}{1 + \cot^2 x}}$
$\cos x$	$\pm \sqrt{1 - \sin^2 x}$	-	$\pm \sqrt{\frac{1}{1 + \tan^2 x}}$	$\pm \sqrt{\frac{\cot^2 x}{1 + \cot^2 x}}$
$\tan x$	$\pm \sqrt{\frac{\sin^2 x}{1 - \sin^2 x}}$	$\pm \sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$	-	$\frac{1}{\cot x}$
$\cot x$	$\pm \sqrt{\frac{1 - \sin^2 x}{\sin^2 x}}$	$\pm \sqrt{\frac{\cos^2 x}{1 - \cos^2 x}}$	$\frac{1}{\tan x}$	-

where the symbol $\pm$ means that the sign depends on the quadrant in which x lies

Addition and subtraction

$$\sin(x_1 \pm x_2) = \sin x_1 \cos x_2 \pm \sin x_2 \cos x_1 \quad (7a)$$

$$\cos(x_1 \pm x_2) = \cos x_1 \cos x_2 \mp \sin x_1 \sin x_2 \quad (7b)$$

$$\tan(x_1 \pm x_2) = \frac{\tan x_1 \pm \tan x_2}{1 \mp \tan x_1 \tan x_2} \quad (8a)$$

$$\cot(x_1 \pm x_2) = \frac{\cot x_1 \cot x_2 \mp 1}{\cot x_2 \pm \cot x_1} \quad (8b)$$

Double-angle formulas

$$\sin(2x) = 2 \sin x \cos x \quad (9a)$$

$$\begin{aligned} \cos(2x) &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 \\ &= 1 - 2 \sin^2 x \end{aligned} \quad (9b)$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x} \quad (10a)$$

$$\cot(2x) = \frac{\cot^2 x - 1}{2 \cot x} \quad (10b)$$

Triple-angle formulas

$$\sin(3x) = 3 \sin x - 4 \sin^3 x \quad (11a)$$

$$\cos(3x) = 4 \cos^3 x - 3 \cos x \quad (11b)$$

Half-angle formulas

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}} \quad (12a)$$

$$\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}} \quad (12b)$$

$$\begin{aligned} \tan \frac{x}{2} &= \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} \\ &= \frac{\sin x}{1 + \cos x} \\ &= \frac{1 - \cos x}{\sin x} \end{aligned} \quad (13a)$$

$$\begin{aligned} \cot \frac{x}{2} &= \pm \sqrt{\frac{1 + \cos x}{1 - \cos x}} \\ &= \frac{\sin x}{1 - \cos x} \\ &= \frac{1 + \cos x}{\sin x} \end{aligned} \quad (14a)$$

Parametric formulas (tangent half-angle substitution)

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \quad (15a)$$

$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \quad (15b)$$

Sum-to-product formulas (prosthaphaeresis)

$$\sin x_1 + \sin x_2 = 2 \sin \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2} \quad (16a)$$

$$\sin x_1 - \sin x_2 = 2 \sin \frac{x_1 - x_2}{2} \cos \frac{x_1 + x_2}{2} \quad (16b)$$

$$\cos x_1 + \cos x_2 = 2 \cos \frac{x_1 + x_2}{2} \cos \frac{x_1 - x_2}{2} \quad (16c)$$

$$\cos x_1 - \cos x_2 = -2 \sin \frac{x_1 + x_2}{2} \sin \frac{x_1 - x_2}{2} \quad (16d)$$

Product-to-sum formulas (Werner's formulas)

$$\sin x_1 \sin x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) - \cos(x_1 + x_2) \right) \quad (17a)$$

$$\sin x_1 \cos x_2 = \frac{1}{2} \left(\sin(x_1 - x_2) + \sin(x_1 + x_2) \right) \quad (17b)$$

$$\cos x_1 \cos x_2 = \frac{1}{2} \left(\cos(x_1 - x_2) + \cos(x_1 + x_2) \right) \quad (17c)$$

Additional formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2} \quad (18a)$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2} \quad (18b)$$

$$\sin x \cos x = \frac{\sin(2x)}{2} \quad (18c)$$